

Ryan Zhenqi Zhou, Ph.D. Candidate

(716) 547-9256 | zhenqizh@buffalo.edu | [Personal Website](#) | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

TECHNICAL SKILLS

- **Programming/Tools:** Python, ArcGIS Pro, QGIS, R, SQL, Git
- **Libraries:** NumPy, Pandas, GeoPandas, Rasterio, SciPy, Scikit-Learn, Keras, TensorFlow, Matplotlib, Seaborn, Bokeh
- **Models:** GeoAI (GWR, MGWR, GRF, SLM, SEM), Regression (OLS), Tree-based Models (RF, XGBoost), Neural Networks (DNN, etc.), Time Series Models (LSTM, Transformer, etc.), Explainable AI (SHAP), Generative AI (GPT)
- **Applications:** Climate, Disaster Resilience, Health Geography, Human Mobility, Accessibility, Economic Geography

WORK EXPERIENCE

Research Assistant, GeoAI Lab, University at Buffalo – SUNY, Buffalo 09/2021 - Present

Climate Disaster Impact Modeling

- Developed data-driven frameworks for assessing **spatial and temporal impacts** of climate disasters using human mobility data, remote sensing images, and 311 service request data
- [GeoAI] Developed **Geographical Random Forests (GRF) in Python** which takes spatial index into consideration [[paper](#)]
- [GeoAI] Applied **time series models (LSTM, Transformer), spatial statistical models (SLM, SEM, GWR, MGWR), and machine learning models (SVM, RF, GRF) with explainable AI (SHAP)** to predict disaster risks and evaluate spatial disparities in physical and social vulnerabilities of communities [[GitHub repository](#)]
- Findings supported local agencies in disaster response and first-author [paper](#) and [paper](#) published and presented at the 2023 and 2025 AAG Conferences

Obesity Prevalence Estimation

- Processed 3GB+ human mobility data and derived **spatial features** of diet and physical activity using **Python**
- [AI] Implemented **models (OLS, GWR, RF, DNN, GRF)** to predict obesity prevalence with accuracy over 90% through cross validation [[GitHub repository](#)]
- First-author [paper](#) published and presented at the 2022 GEOMED Conference

Geo-Knowledge-Guided GPT Model

- [AI] Fused geo-knowledge with **GPT models (GPT-2 to GPT-4)** via prompt engineering and question-answering to extract and classify complex location descriptions
- Achieved over 40% improvement in accuracy compared to NER baselines [[paper](#)]

Geospatial Data Scientist | Consultant, World Bank - IFC, Washington, D.C.

05/2024 - Present

Geospatial Analysis and Modeling in Healthcare, Climate, and Tourism

- Processed, analyzed, and mapped **large-scale geospatial datasets**, including healthcare facility data, patient records, climate disaster data (floods, extreme heat, and droughts), tourism data, wealth index data, population data, and remote sensing images in low- and middle-income countries (LMICs) using **Python, ArcGIS Pro, and Map API**
- Conducted **geospatial analysis** of 1,000+ healthcare facilities in Kenya and Tanzania, linking quality scores with climate and economic data to generate outcome indicators for resilience and equity
- [GeoAI] Built predictive models: **Random Forest and Linear Regression** for wealth and population estimation (95%+ accuracy). **XGBoost and Huff Model** for forecasting healthcare facility transactions and market demand. Developed **GPT-based address-matching pipeline** to standardize unstructured data
- Results informed resilience-focused reforms and were adopted by local providers to improve healthcare service delivery

Research Assistant, Weill Cornell Medical College, Cornell University, New York City

08/2024 - 06/2025

Geospatial Evaluation of Cardiovascular Disease Risk Factors

- Collected and processed **large geospatial datasets** on cardiovascular disease, socioeconomic status, and demographic at the neighborhood level in New York City
- Applied fuzzy string matching to **geocode** patient data and aggregate it to neighborhood level for analysis and modeling
- Built **Python** pipelines for scalable and reproducible data processing workflows [[GitHub repository](#)]

EDUCATION

University at Buffalo – SUNY, USA

9/2021 - 09/2026 (expected)

Ph.D. in GeoAI and Geospatial Data Science (STEM)

- **Geographic Information System Coursework:** GeoAI, Geospatial Big Data Analytics, Remote Sensing, Cartography
- **Data Science Coursework:** Machine Learning, Deep Learning, Time Series Modeling, Database, Data Visualization